Discussion Section 6
STAT 132 , Winter 2025

1. Suppose that five random variables $X_1, \ldots, X_5$ are independently sampled from a standard normal distribution. Find a value for the constant c such that the random variable

$$U = \frac{c(X_1 + X_2)}{\sqrt{X_3^2 + X_4^2 + X_5^2}}$$

will have a t distribution.

2. Suppose that $X_1, \ldots, X_n$ form a random sample from the normal distribution with unknown mean μ and known variance σ^2. Let $\Phi(\cdot)$ stand for the CDF of the standard normal distribution, and let $\Phi^{-1}(\cdot)$ be its inverse. Show that the following interval is a $1 - \alpha = \gamma$ confidence interval for μ:

$$\left(\bar{X} - \Phi^{-1}\left(\frac{1+\gamma}{2}\right)\frac{\sigma}{\sqrt{n}}, \bar{X} + \Phi^{-1}\left(\frac{1+\gamma}{2}\right)\frac{\sigma}{\sqrt{n}}\right)$$

3. Suppose that $X_1, \ldots, X_n$ form a random sample from the normal distribution with unknown mean μ and known variance σ^2. How large a random sample must be taken in order to be sure that a 95% confidence interval for μ will have length less than 0.01σ ?

4. Suppose that $X_1, \ldots, X_n$ form a random sample from the exponential distribution with unknown mean μ. Describe a method for constructing a $1 - \alpha$ confidence interval for μ. *Hint:* Use $\frac{1}{\mu}\sum_{i=1}^{n} X_i$ as a pivotal quantity.